الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني

الامتحانات الرسمية لشهادة الامتياز الفني
لدورة عام 2018 الثانية

الاسم والشهرة : .....
رقم المرشح : .....

الاختصاص والرمز : الالكترونيك (20203)
المادة : الالكترونيك الرقمي
المستندات المسموح بها : آلة حاسبة غير مبرمجة

المعدل : (10)
المدة : ساعتان

Traiter quatre questions seulement: (5 pts /question)

I- On veut réaliser un compteur décimal de 00 à 99. Pour ceci, on utilise un microprocesseur 6809 avec ses circuits d'horloge et d'alimentation, un PIA 6821, deux afficheurs 7 segments avec leurs décodeurs BCD/7segments convenables et des résistances.
- a- Dessiner le circuit complet en choisissant les entrées et sorties.
- b- Conformément à ce circuit, écrire le programme en langage assembleur que doit exécuter le microprocesseur.

II- Une liste d'octets signés débute à l'adresse $000E et dont la longueur est le contenu de la mémoire $000D. Ecrire un programme en langage assembleur 6809 pour calculer le nombre des éléments pairs de cette liste et le placer dans l'adresse $000C puis le nombre des éléments impairs de cette liste et le placer dans l'adresse $000B.

III- Considérer le programme de la figure 5:
- a- Réécrire ce programme et commenter l'opération de chaque instruction.
- b- Donner l'objectif de ce programme.

IV- Ecrire, en langage assembleur PIC16F84, le programme qui fait sortir sur le PORTB et avec un retard de 10ms produit par TIMER0, les octets de la liste débutant à l'adresse 0X20 et se terminant à l'adresse 0X3D. Le quartz utilisé étant de 12MHz, calculer et remplir un tableau comme suit :

Quartz	Durée du cycle	Prédiviseur PS2:PS0	TMR0 (decimal)
12 MHz			

V- Répondre aux questions suivantes:
- a- Compléter la phrase suivante: "L'activation du bouton dans la figure 1 effectue le ... du microcontrôleur".
- b- Quelle est la différence entre les figures 1 et 2?
- c- Donner l'opération de la séquence représentée par la figure 4.
- d- Donner le rôle de chaque bit du registre INTCON du PIC16F84.
- e- Compléter la phrase suivante: "La figure 3 représente le fonctionnement de ...".

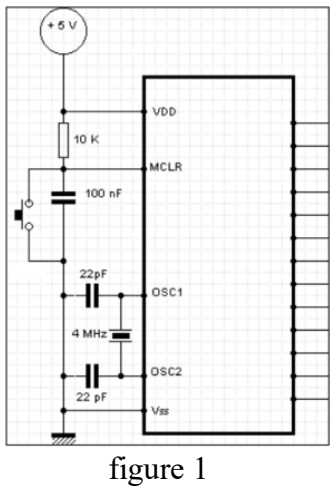


figure 1

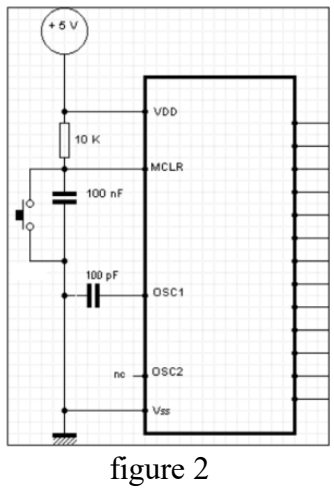


figure 2

```
#INCLUDE <P16F84.INC>
SHIFT     EQU     0X0C
COMPT1    EQU     0X0D
          ORG     0X00
          GOTO    START
          ORG     0X04
          GOTO    INT_VECT
START     ORG     0X05
          BSF     STATUS,RP0
          CLRF    TRISB
          MOVLW   B'00000100'
          MOVWF   OPTION_REG
          BCF     STATUS,RP0
          MOVLW   D'125'
          MOVWF   COMPT1
          MOVLW   B'10100000'
          MOVWF   INTCON
LOOP1     GOTO    LOOP1
INT_VECT  MOVLW   B'00000001'
          MOVWF   SHIFT
          MOVF    SHIFT,0
          MOVWF   PORTB
          BCF     STATUS,C
          RLF     SHIFT,1
          BTFSC   SHIFT,4
          SWAPF   SHIFT,F
          CALL    DELAY
DELAY     MOVLW   D'6'
          MOVWF   TMR0
TEST      MOVF    TMR0,W
          BTFSS   STATUS,Z
          GOTO    TEST
          MOVLW   D'6'
          MOVWF   TMR0
          DECFSZ  COMPT1,1
          GOTO    TEST
          MOVLW   D'6'
          MOVWF   TMR0
          MOVLW   D'125'
          MOVWF   COMPT1
          RETURN
          END
```
figure 5

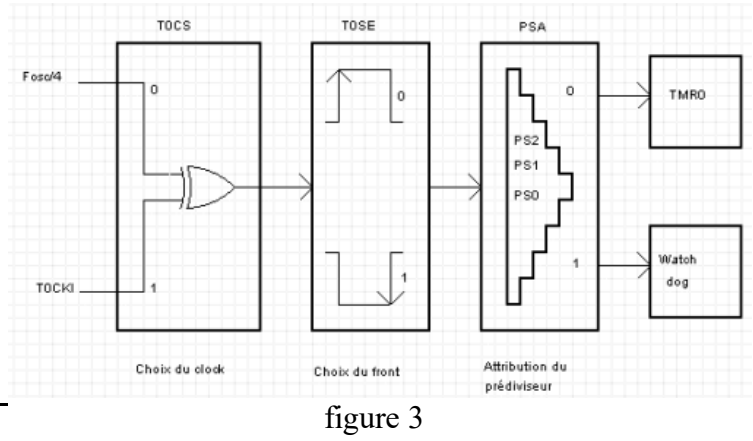


figure 3

```
BCF     STATUS,RP0
MOVLW   adresse dont on veut lire le contenu
MOVWF   EEADR
BSF     STATUS,RP0
BCF     EECON1, EPGD
BSF     EECON1, RD
BCF     STATUS,RP0
MOVF    EEDATA,W
```
figure 4

ANNEXE / APPENDIX

Instruction set of 6809

ADCA, ADCB, ADDA, ADDB, ADDD, ANDA, ANDB, ANDCC, ASLA, ASLB, ASL, ASRA, ASRB, ASR, BITA, BITB, CLRA, CLRB, CLR, CMPA, CMPB, CMPD, CMPU, CMPS, CMPX, CMPY, COMA, COMB, COM, CWAI, DAA, DECA, DECB, DEC, EORA, EORB, EXG, INCA, INCB, INC, JMP, JSR, LDA, LDB, LDD, LDS, LDU, LDX, LDY, LEAS, LEAU, LEAX, LEAY, LSLA, LSLB, LSL, LSRA, LSRB, LSR, MUL, NEGA, NEGB, NEG, NOP, ORA, ORB, ORCC, PSHS, PSHU, PULS, PULU, ROLA, ROLB, ROL, RORA, RORB, ROR, RTI, RTS, SBCA, SBCB, SEX, STA, STB, STD, STS, STU, STX, STY, SUBA, SUBB, SUBD, SWI, SWI2, SWI3, SYNC, TFR, TSTA, TSTB, TST.

PIA:

ORA et DDRA = $8000	**CRA = $8001**
ORB et DDRB = $8002	**CRB = $8003**

Instruction set of PIC 16F84A

ADDWF, ANDWF, CLRF, CLRW, COMF, DECF, DECFSZ, INCF, INCFSZ, IORWF, MOVF, MOVWF, NOP, RLF, RRF, SUBWF, SWAPF, XORWF, BCF, BSF, BTFSC, BTFSS, ADDLW, ANDLW, CALL, CLRWDT, GOTO, IORLW, MOVLW, RETFIE, RETLW, RETURN, SLEEP, SUBLW, XORLW.

Internal architecture of PIC16F84A

File Address	Bank 0	Bank 1	File Address
00h	Indirect addr.(1)	Indirect addr.(1)	80h
01h	TMR0	OPTION_REG	81h
02h	PCL	PCL	82h
03h	STATUS	STATUS	83h
04h	FSR	FSR	84h
05h	PORTA	TRISA	85h
06h	PORTB	TRISB	86h
07h	—	—	87h
08h	EEDATA	EECON1	88h
09h	EEADR	EECON2(1)	89h
0Ah	PCLATH	PCLATH	8Ah
0Bh	INTCON	INTCON	8Bh
0Ch –	General Purpose Registers (SRAM)	Mapped (accesses) in Bank 0	8Ch –
			FFh

PS2:PS0: Prescaler Rate Select bits

Bit Value	TMR0 Rate	WDT Rate
000	1 : 2	1 : 1
001	1 : 4	1 : 2
010	1 : 8	1 : 4
011	1 : 16	1 : 8
100	1 : 32	1 : 16
101	1 : 64	1 : 32
110	1 : 128	1 : 64
111	1 : 256	1 : 128

INTCON

GIE	EEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF

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Treat four questions only: (5 pts /question)

I- We want to design a decimal counter from 00 to 99. For this, we use a microprocessor 6809 with its clock and supply circuits, a PIA 6821, two 7segment display with their appropriate BCD/7segment decoders and resistors.
- a- Draw the complete circuit by choosing inputs and outputs.
- b- According to this circuit, write the assembly language program that the microprocessor must execute.

II- A list of signed bytes starts at address $000E and which length is the content of the memory $000D. Write a program in assembly language 6809 to calculate the number of odd elements in this list and place it in the address $000C then the number of even elements in this list and place it in the address $000B.

III- Consider the program in figure 5:
- a- Rewrite the program and comment the operation of each instruction.
- b- Give the aim of this program.

IV- Write, in assembly language PIC16F84, the program that outputs at PORTB and with a delay of 10ms produced by TIMER0, the bytes of the list starting at address 0X20 and ending at address 0X3D. The used quartz being of 12MHz, calculate and fill in a table as follow:

Quartz	Cycle duration	Prescaler PS2:PS0	TMR0 (decimal)
12 MHz			

V- Answer the following questions :
- a- Complete the following phrase: "Pushing the button in figure 1 makes the ... of the microcontroller".
- b- What is the difference between figures 1 and 2?
- c- Give the operation of the sequence in figure 4.
- d- Give the role of each bit of the INTCON register in PIC16F84.
- e- Complete the following phrase: "Figure 3 represents the operation of ...".

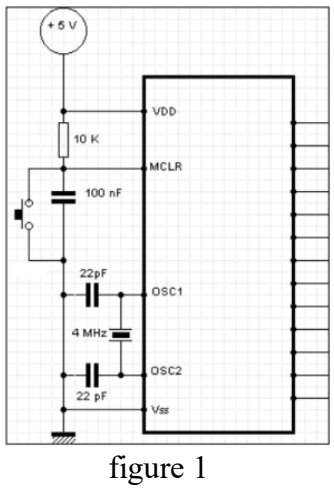


figure 1

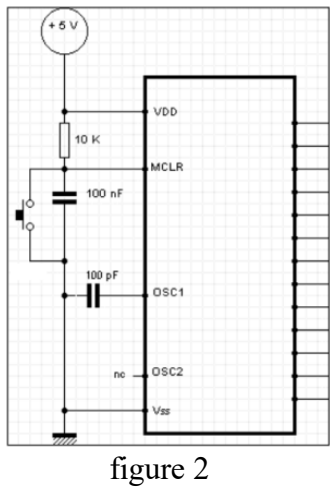


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START     ORG     0X05
          BSF     STATUS,RP0
          CLRF    TRISB
          MOVLW   B'00000100'
          MOVWF   OPTION_REG
          BCF     STATUS,RP0
          MOVLW   D'125'
          MOVWF   COMPT1
          MOVLW   B'10100000'
          MOVWF   INTCON
LOOP1     GOTO    LOOP1
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          MOVWF   SHIFT
          MOVF    SHIFT,0
          MOVWF   PORTB
          BCF     STATUS,C
          RLF     SHIFT,1
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          SWAPF   SHIFT,F
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          GOTO    TEST
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          MOVLW   D'125'
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          RETURN
          END
```
figure 5

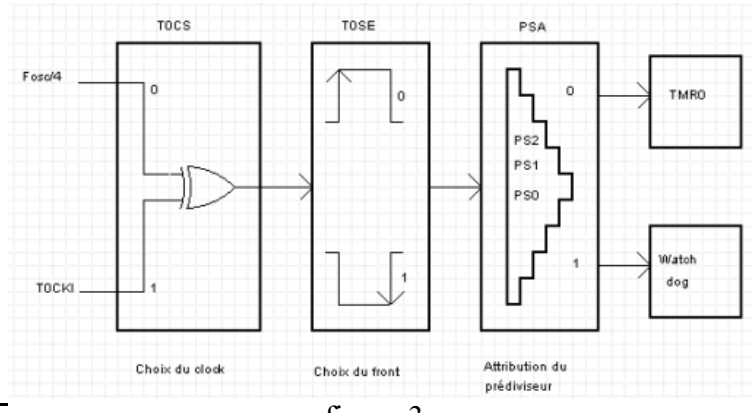


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